

Accumulator shut-off block

Type ABZSS, 0532VAW

RE50129-B/03.10
Material no. R901268180

English

Operating instructions



Type ABZSS



Type 0532VAW

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Contents

1	About this document	6
1.1	Related documents	6
1.2	Abbreviations used	6
2	General safety instructions	7
2.1	Intended use	7
2.2	Improper use	7
2.3	Qualification of personnel	8
2.4	Safety instructions in this document	8
2.5	Adhere to the following instructions	9
2.6	Obligations of the machine manufacturer	10
2.7	Obligations of the operator	10
3	Scope of delivery	11
4	Product description	12
4.1	Performance description	12
4.2	Device description	12
4.3	Interfaces	17
4.4	Sizes	22
4.5	Operating elements	22
4.6	Manifold identification	24
5	Transport and storage	26
5.1	Transporting the manifold	26
5.2	Storing the manifold	28
6	Assembly	29
6.1	Unpacking	29
6.2	Installation conditions	29
6.3	Assembling the manifold	30
6.4	Assembly of the pressure relief valve	34
7	Commissioning	35
7.1	First commissioning	35
7.2	Re-commissioning after extended standstill	37
8	Operation	38
9	Maintenance and repair	39
9.1	Cleaning and care (maintenance)	39
9.2	Maintenance plan	39
9.3	Maintenance	40
9.4	Repair	41
9.5	Spare parts	41
10	Decommissioning	42
11	Disassembly and replacement	42
11.1	Required tools	42
11.2	Preparing for disassembly	42
11.3	Disassembling the manifold	42
11.4	Preparing the manifold for storage/further use	43
11.5	Replacing the manifold	43
12	Disposal	44
12.1	Environmental protection	44
13	Extension and conversion	44
14	Troubleshooting	45
14.1	How to proceed for troubleshooting	45
15	Technical data	47
16	Appendix	48
16.1	Address directory	48

About this document

1 About this document

These instructions contain important information on the safe and appropriate assembly, transport, commissioning, operation, maintenance, disassembly and simple troubleshooting of the accumulator shut-off block, afterwards referred to as “manifold”.

- Read these instructions completely, especially chapter 2 “General safety instructions” before working with the manifold.

Instructions for use Apart from the instructions in chapter 2 that must always be observed, each main chapter can be consulted separately; thus, the safety instructions are repeated in the respective main chapter.

1.1 Related documents

- Please observe the following documentation in addition:
 - System documentation from the machine manufacturer
 - Operating instructions of the hydraulic accumulator manufacturers
 - Operating instructions of the attached components
 - Data sheet RE 50131 “Accumulator shut-off block ABZSS”
 - Data sheet RE 50128 “Accumulator shut-off block 0532VAW”
 - Data sheet RE 90220 “Hydraulic fluids on a mineral oil basis and related hydrocarbons”

You can find these operating instructions as well as the related documents in the media directory under www.boschrexroth.com/variou utilities/mediadirectory/

- Also observe the generally applicable, legal or otherwise binding regulations of the European or national legislation and the rules for the prevention of accidents and for environmental protection applicable in your country.

1.2 Abbreviations used

Table 1: Abbreviations

Abbreviation	Meaning
RE	Rexroth document in English
Rx	Rexroth document in another language
REXXXXX-B	Rexroth operating instructions in English
RXXXXXX-B	Rexroth operating instructions in another language

2 General safety instructions

The manifold has been manufactured according to the accepted rules of current technology. There is, however, still a risk of personal injury or damage to property if the following safety instructions and warnings before instructions contained in these operating instructions are not observed.

- ▶ Read these instructions completely and thoroughly before working with the manifold.
- ▶ Keep these instructions in a location where they are accessible to all users at all times.
- ▶ Always include the operating instructions when you pass the manifold on to third parties.

2.1 Intended use

The manifold is used for safety, isolating and unloading functions in hydraulic accumulators. It meets the requirements and safety regulations according to DIN 24552 Pressure vessels regulation and Technical rules (TRB 403 and/or TRB 404). The manifold may only be operated within the admissible operating conditions and performance limits.

In terms of the Pressure Equipment Directive 93/27/EC, the manifold is an assembly in the sense of article 2.1.5. According to article 3, paragraph 3, the manifold must not bear a CE mark in the sense of 2.1.5.

The manifold is intended exclusively for integration into a machine or system or to be assembled with other components to form a machine or system.

The manifold may only be commissioned if it has been integrated in the machine/system for which it is designed and the machine/system fully complies with the EU Machinery Directive.



The manifold is not considered to be a safety component in terms of EC Machinery Directive 2006/42/EC.

- ▶ Observe the operating conditions and performance limits specified in chapter 15 "Technical data".

The manifold is a work appliance and not designed for private use.

Intended use includes having completely read and understood these instructions, especially the chapter 2 "General safety instructions".

2.2 Improper use

Any use of the manifold other than described in chapter "Intended use" is considered as improper.

The manifold with the type tested pressure relief valve must not accept any control tasks. The relevant performance limits of the installed components must not be exceeded.

General safety instructions

2.3 Qualification of personnel

Assembly, commissioning, disassembly and maintenance (incl. servicing and care) require a basic knowledge of mechanics and hydraulics, as well as familiarity with the associated technical terms. In order to ensure operational safety, these activities may only be carried out by corresponding experts or an instructed person under the direction and supervision of an expert.

Experts are those who can recognize potential hazards and apply the appropriate safety measures due to their professional training, knowledge and experience, as well as their understanding of the relevant conditions pertaining to the work to be undertaken. An expert must observe the rules/laws relevant to the specific subject area.

Hydraulic expert knowledge means, e.g. that the personnel are able to read and completely understand hydraulic schemes. Particularly the functionality of the installed safety components in the overall safety concept must be completely understood.

2.4 Safety instructions in this document

These instructions contain warning signs followed by an instruction whenever there is a risk of personal injury or damage to the equipment. The measures described for preventing these hazards must be observed.

Safety instructions are set out as follows:

SIGNAL WORD	Type of risk
	Consequences ► Precautions

- **Warning sign (warning triangle):** Draws attention to the hazard
- **Signal word:** Identifies the degree of hazard
- **Type of risk:** Specifies the type or source of the hazard
- **Consequences:** Describes the consequences of non-compliance
- **Precautions:** Specifies how the hazard can be prevented

The signal words have the following meaning:

Table 2: Meaning signal words

Signal word	Application
DANGER! 	Indicates an imminently hazardous situation which, if not avoided, will certainly result in death or serious injury.
WARNING! 	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
CAUTION! 	Indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury or damage to equipment.
	If this information is disregarded, the operating procedure may be impaired.

2.5 Adhere to the following instructions

2.5.1 General instructions

- Observe the regulations for accident prevention and environmental protection for the country where the product is used and at the workplace.
- Exclusively use Rexroth products in good technical order and condition.
- Do not modify or retrofit the manifold.
- Only use the product within the performance range provided in the technical data.
- Persons who assemble, operate, disassemble or maintain Rexroth products must not consume any alcohol, drugs or pharmaceuticals that may affect their ability to respond.
- Make sure that all safety equipment belonging to the product is present, has been installed properly and is fully functional. Do not displace, bypass or disable the safety equipment.
- Before starting any work at the manifold, make yourself familiar with the position of the closest emergency stop equipment for stopping the system.
- If it should be necessary to disable any safety equipment temporarily, for example for commissioning or maintenance work, always take the appropriate measures to ensure that no hazard to a person's life or health or to property may occur. Also observe the superordinate operating instructions for the machine or system.
- The manifold considerably heats up during operation. Allow the manifold to cool down sufficiently before touching it. Wear heat-protective clothing, e.g. gloves.
- The warranty only applies to the delivered configuration. The warranty will not apply if the product is incorrectly assembled, not used as intended and/or handled or maintained improperly.
- Any mechanical modification like e.g. drilling, milling, cutting, etc. is forbidden.
- Do not expose the product to any mechanical loads under any circumstances. Never use the product as a handle or step. Do not place any objects on top of the manifold.

General safety instructions

2.6 Obligations of the machine manufacturer

In order to ensure safety when handling the manifold, the manufacturer of the machine must:

- Select the safest installation position for the manifold possible in order to avoid possible damage of the manifold and its add-on units.
- Provide for sufficient protection of the manifold including all add-on units depending on the selected installation position in order to avoid the possible damage of the manifold and its add-on units.
- Prescribe the wearing of personal protective equipment or perform suitable protection measures in order to reduce the risk of collision, the risk of hearing damage and the risk of burns at hot surfaces for operating and maintenance personnel.
- Label the hydraulic components according to the specifications of EN 982, section 7.3 et seq.
- Take suitable protective measures in order to reduce the slip hazard caused by oily surfaces around the manifold.

2.7 Obligations of the operator

The operator of the machine into which the manifold is integrated is bound to provide for personnel training on a regular basis regarding the following subjects:

- Observation and use of the operating instructions as well as the legal stipulations
- Intended use of the manifold.
- Observation of the operating instructions from the machine manufacturer and operator, and the factory security offices, if applicable.
- What to do in an emergency

Corresponding proof of implementation should be documented.



Bosch Rexroth offers training support in specific fields. You can find an overview of the training contents in the Internet under <http://www.boschrexroth.de/didactic>.

3 Scope of delivery

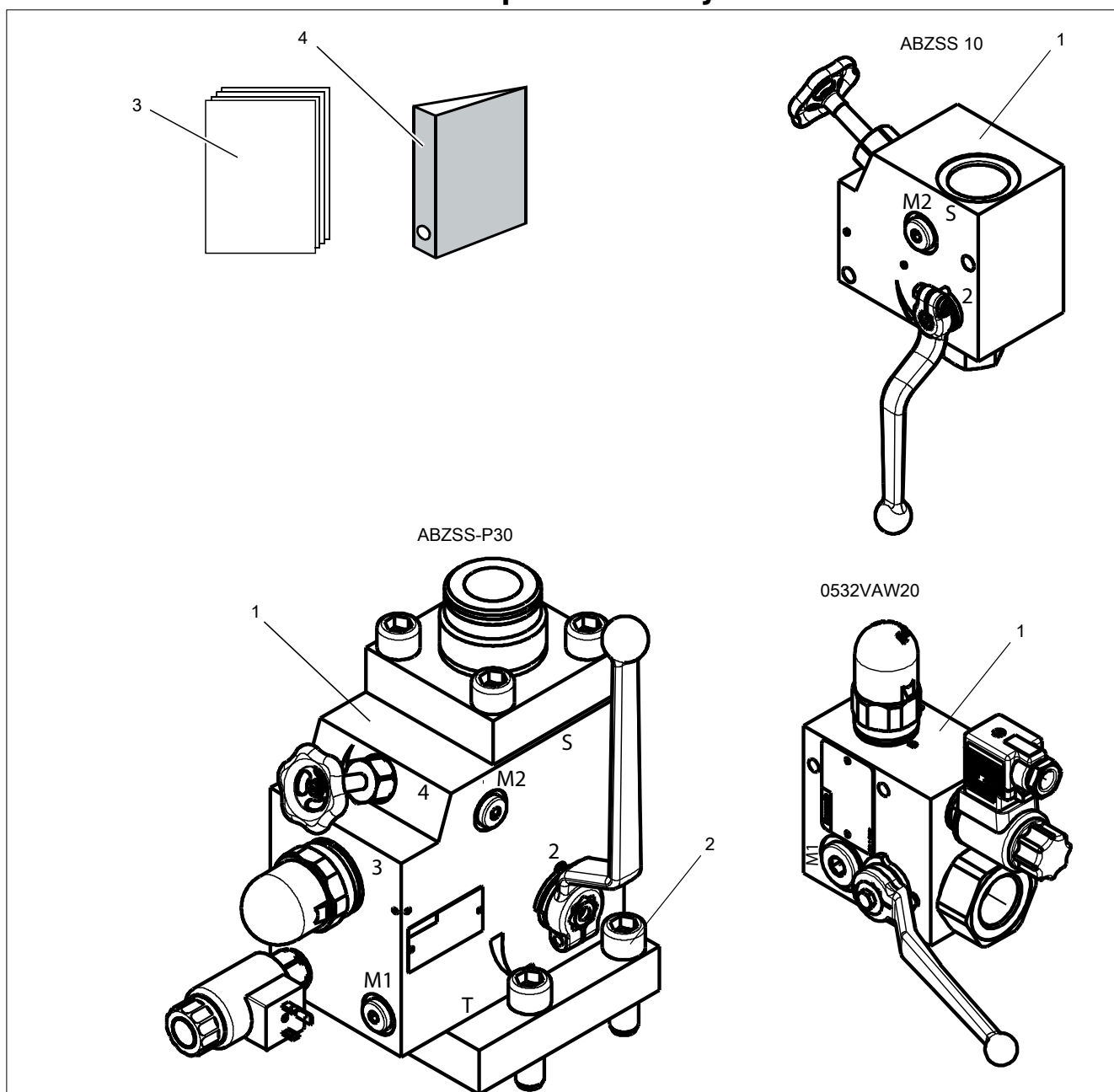


Fig. 1: Scope of delivery accumulator shut-off block ABZSS, ABZSS-P30 and 0532VAW

The scope of delivery includes:

- 1x accumulator shut-off block with or without pressure relief valve (depending on the version) (1)
- 4x fastening screws (only with type ABZSS-P30) (2)
- 1x test certificate of the type tested pressure relief valve (only in versions with pressure relief valve) (3)
- 1x operating instructions accumulator shut-off block (4)

4 Product description

4.1 Performance description

By means of the manifold, the accumulator is protected from inadmissible overpressure. If the maximum operating pressure is exceeded, the pressure relief valve opens and vents the overpressure. The pressure relief valve must not accept any control function. It exclusively functions as relief valve.

4.2 Device description

The manifold is used for safety, isolating and unloading functions in hydraulic accumulators.

It meets the requirements and safety regulations according to the Technical rules for pressure containers (TRB 403 and/or TRB 404).

The connection between the manifold and the accumulator is established by means of an accumulator adapter. An optional, additional electrically operated 2 way valve (normally open) allows for the automatic unloading of the accumulator upon shut-down or "emergency stop function".

By means of the pressure relief valve, the accumulator is protected from inadmissible overpressure. The **pressure relief valve must not assume control tasks!**

Care must be taken that the distance between the pressure set at the pressure relief valve and the operating pressure is sufficient. Responding of the pressure relief valve is to be prevented as far as possible.

Product description

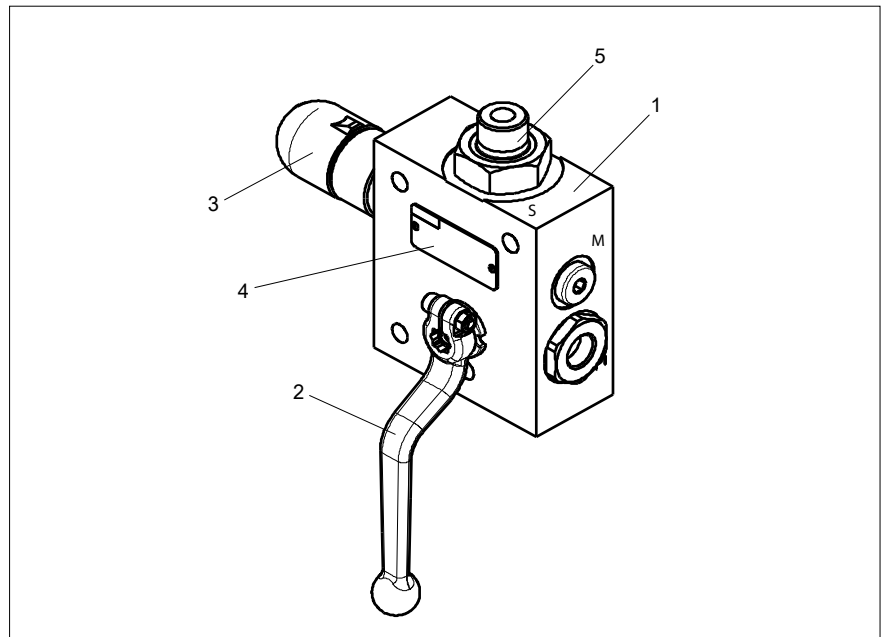


Fig. 2: Accumulator shut-off block ABZSS08

- | | | | |
|---|--|---|-------------------------------|
| 1 | Block | 4 | Name plate |
| 2 | System shut-off cock with manual unloading of the pressure | 5 | Accumulator adapter, optional |
| 3 | Pressure relief valve, optional | | |

Block (1)	Manifold is used for accepting the components and/or valves
System shut-off cock with manual unloading of the pressure (2)	The system shut-off cock serves the isolation of the pressure line and the manual unloading of the accumulator
Pressure relief valve (3)	Pressure relief valve protects the hydraulic accumulator against inadmissible overpressure
Name plate (4)	Name plate for the unambiguous identification of the manifold
Accumulator adapter (5)	Establishes the connection between manifold and accumulator

Product description

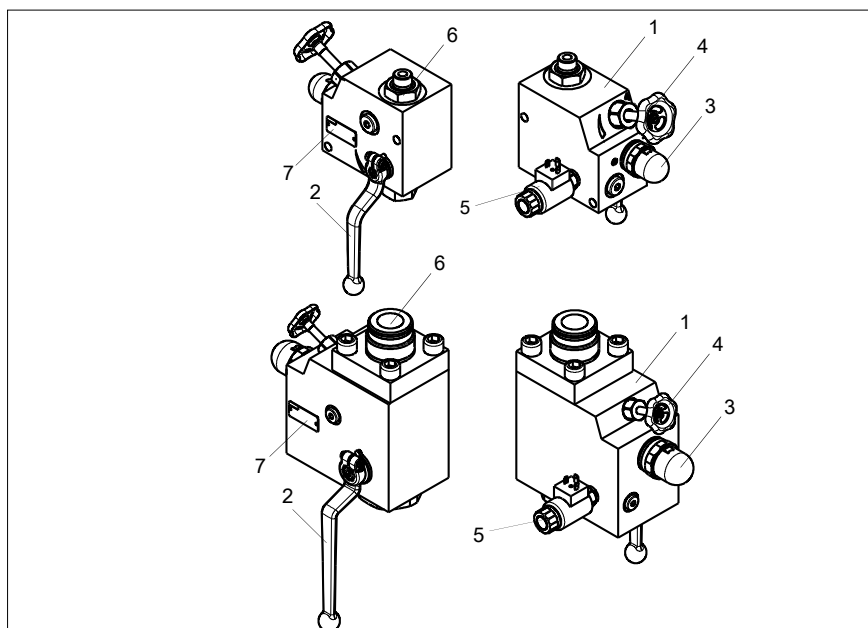


Abb. 3: Accumulator shut-off block ABZSS10, ABZSS20 and ABZSS30

- | | |
|-----------------------------------|--|
| 1 Block | 5 Electro-magnetic unloading, optional |
| 2 System shut-off cock | 6 Accumulator adapter, optional |
| 3 Pressure relief valve, optional | 7 Name plate |
| 4 Manual unloading | |

- | | |
|---------------------------------------|--|
| Block (1) | Manifold is used for accepting the components and/or valves |
| System shut-off cock (2) | System shut-off cock serves the isolation of the pressure line |
| Pressure relief valve (3) | Pressure relief valve protects the hydraulic accumulator against inadmissible overpressure |
| Manual unloading (4) | Manual unloading, this valve is used for the manual unloading of the accumulator |
| Electro-magnetic unloading (5) | Electro-magnetic unloading, this valve can be used for the electro-magnetic unloading of the accumulator |
| Accumulator adapter (6) | Establishes the connection between manifold and accumulator |
| Name plate (7) | Name plate for the unambiguous identification of the manifold |

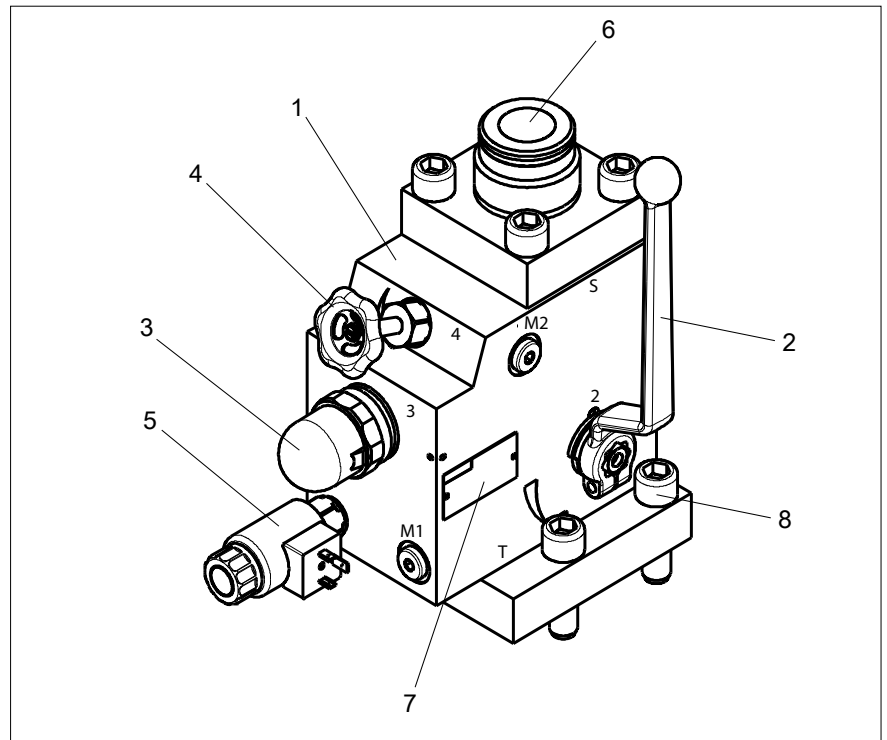
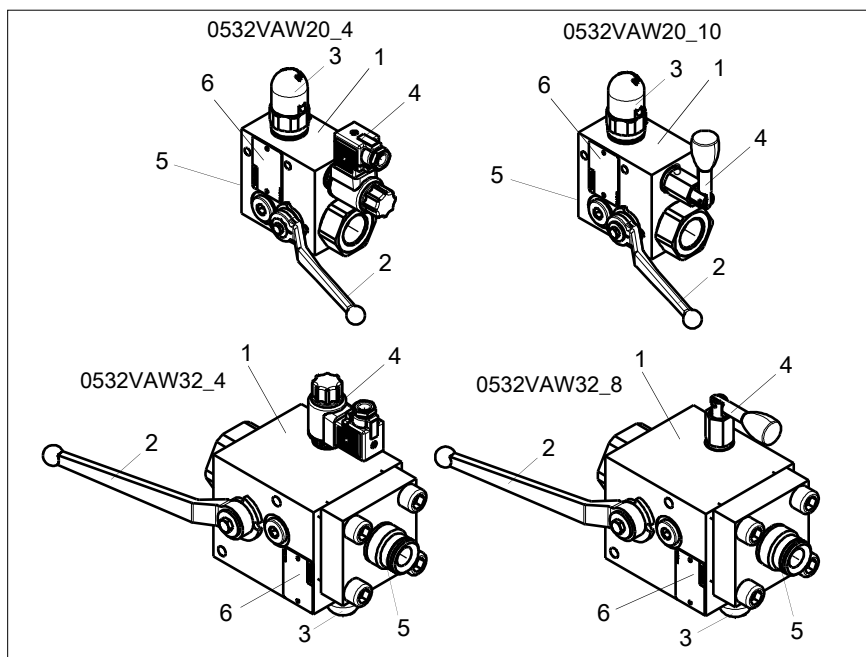


Fig. 4: Accumulator shut-off block ABZSS-P30

- | | | | |
|---|---------------------------------|---|--------------------------------------|
| 1 | Block | 5 | Electro-magnetic unloading, optional |
| 2 | System shut-off cock | 6 | Accumulator adapter, optional |
| 3 | Pressure relief valve, optional | 7 | Name plate |
| 4 | Manual unloading | 8 | 4 fastening screws |

- | | |
|---------------------------------------|--|
| Block (1) | Manifold is used for accepting the components and/or valves |
| System shut-off cock (2) | System shut-off cock serves the isolation of the pressure line |
| Pressure relief valve (3) | Pressure relief valve protects the hydraulic accumulator against inadmissible overpressure |
| Manual unloading (4) | Manual unloading, this valve is used for the manual unloading of the accumulator |
| Electro-magnetic unloading (5) | Electro-magnetic unloading, this valve can be used for the electro-magnetic unloading of the accumulator |
| Accumulator adapter (6) | Establishes the connection between manifold and accumulator |
| Name plate (7) | Name plate for the unambiguous identification of the manifold |

Product description**Fig. 5: Accumulator shut-off block 0532VAW20 and 0532VAW32**

- | | |
|---|--|
| 1 Block | 5 Accumulator adapter, optional |
| 2 System shut-off cock | 6 Name plate |
| 3 Pressure relief valve, optional | |
| 4 Manual or electro-magnetic unloading, optional | |

Block (1)	Manifold is used for accepting the components and/or valves
System shut-off cock (2)	System shut-off cock serves the isolation of the pressure line
Pressure relief valve (3)	Pressure relief valve protects the hydraulic accumulator against inadmissible overpressure
Manual unloading (4)	Manual unloading, this valve is used for the manual unloading of the accumulator
Electro-magnetic unloading (4)	Electro-magnetic unloading, this valve can be used for the electro-magnetic unloading of the accumulator
Accumulator adapter (5)	Establishes the connection between manifold and accumulator
Name plate (6)	Name plate for the unambiguous identification of the manifold

4.3 Interfaces

4.3.1 Hydraulic connections

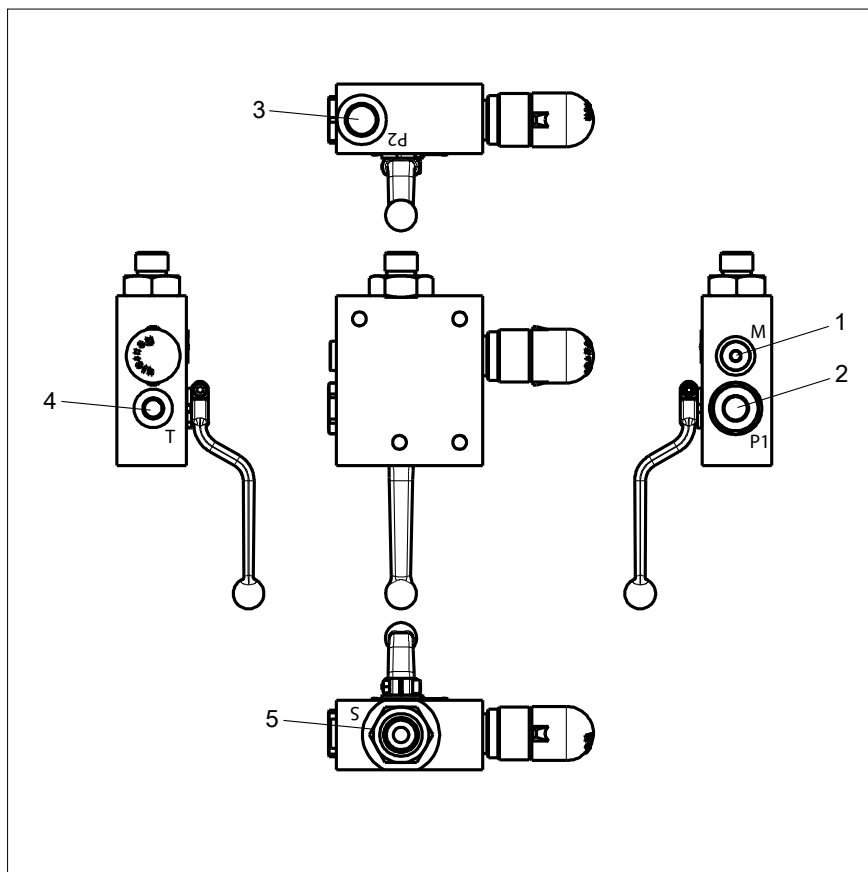


Fig. 6: Hydraulic connections ABZSS 08

- | | |
|------------------------|------------------------|
| 1 Measuring port "M" | 4 Tank port "T" |
| 2 Pump connection "P1" | 5 Accumulator port "S" |
| 3 Pump connection "P2" | |

- | | |
|---------------------------------|--|
| Measuring port "M" (1) | Measuring port for the pressure monitoring (see circuit diagram). |
| Pump connection "P1" (2) | Pump connection for filling the accumulators with oil. |
| Pump connection "P2" (3) | Pump connection for filling the accumulators with oil. |
| Tank port "T" (4) | Via the tank port, the oil is led back to the tank (discharge line) in case of inadmissible overpressure or manual unloading of the accumulator. |
| Accumulator port "S" (5) | Via the accumulator port, the hydraulic accumulator is connected to the manifold. |

Product description

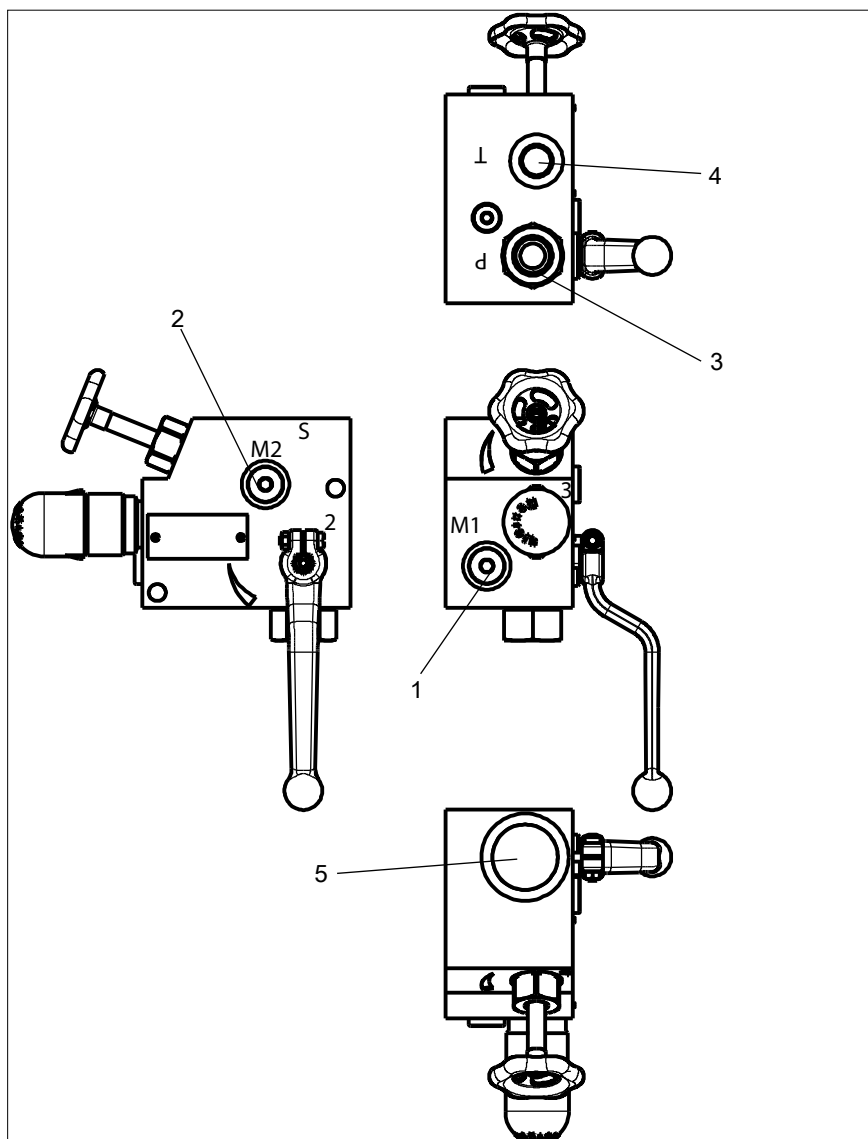


Fig. 7: Accumulator shut-off block ABZSS10, ABZSS20 and ABZSS30

- | | |
|-----------------------|------------------------|
| 1 Measuring port "M1" | 4 Tank port "T" |
| 2 Measuring port "M2" | 5 Accumulator port "S" |
| 3 Pump connection "P" | |

- | | |
|---------------------------------|--|
| Measuring port "M1" (1) | Measuring port for the pressure monitoring (see circuit diagram). |
| Measuring port "M2" (2) | Measuring port for the pressure monitoring (see circuit diagram). |
| Pump connection "P" (3) | Pump connection for filling the accumulators with oil. |
| Tank port "T" (4) | Via the tank port, the oil is led back to the tank (discharge line) in case of inadmissible overpressure or electric unloading of the accumulator. |
| Accumulator port "S" (5) | Via the accumulator port, the hydraulic accumulator is connected to the manifold. |

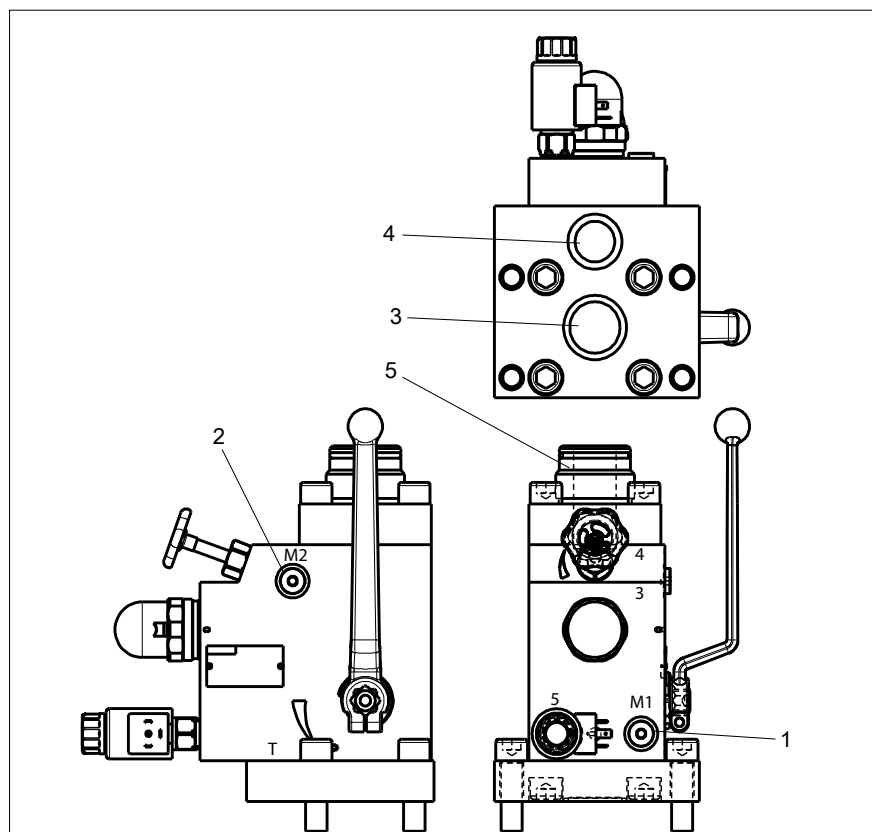


Fig. 8: Accumulator shut-off block ABZSS-P30

- | | | | |
|---|---------------------|---|----------|
| 1 | Measuring port "M1" | 4 | Port "T" |
| 2 | Measuring port "M2" | 5 | Port "S" |
| 3 | Pump connection "P" | | |

- | | |
|---------------------------------|--|
| Measuring port "M1" (1) | Measuring port for the pressure monitoring (see circuit diagram). |
| Measuring port "M2" (2) | Measuring port for the pressure monitoring (see circuit diagram). |
| Pump connection "P" (3) | Pump connection for filling the accumulators with oil. |
| Tank port "T" (4) | Via the tank port, the oil is led back to the tank (discharge line) in case of inadmissible overpressure or electric unloading of the accumulator. |
| Accumulator port "S" (5) | Via the accumulator port, the hydraulic accumulator is connected to the manifold. |

Product description

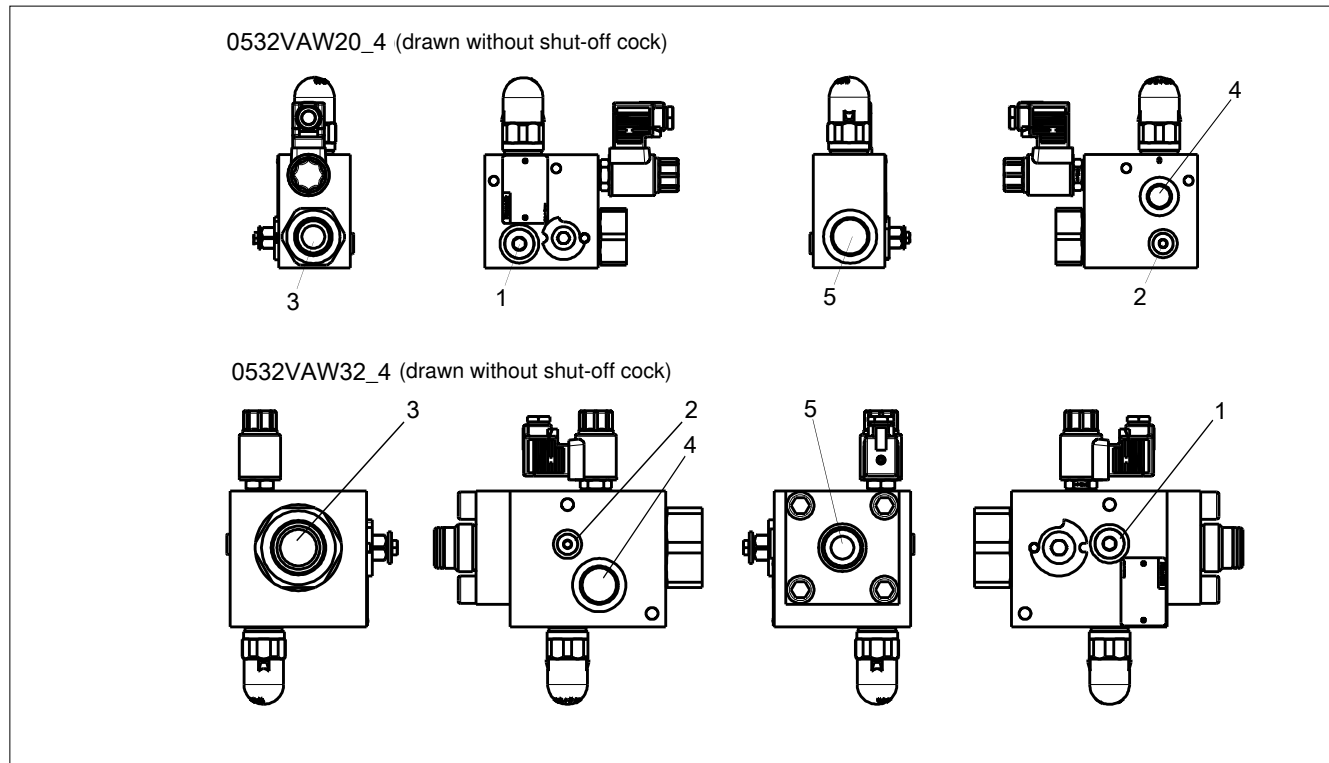


Fig. 9: Accumulator shut-off block 0532VAW20 and 0532VAW32

- | | |
|------------------------------|-------------------------------|
| 1 Measuring port "M1" | 4 Tank port "T" |
| 2 Measuring port "M2" | 5 Accumulator port "S" |
| 3 Pump connection "P" | |

- | | |
|---------------------------------|--|
| Measuring port "M1" (1) | Measuring port for the pressure monitoring (see circuit diagram). |
| Measuring port "M2" (2) | Measuring port for the pressure monitoring (see circuit diagram). |
| Pump connection "P" (3) | Pump connection for filling the accumulators with oil. |
| Tank port "T" (4) | Via the tank port, the oil is led back to the tank (discharge line) in case of inadmissible overpressure or electric unloading of the accumulator. |
| Accumulator port "S" (5) | Via the accumulator port, the hydraulic accumulator is connected to the manifold. |

4.3.2 Electrical connections

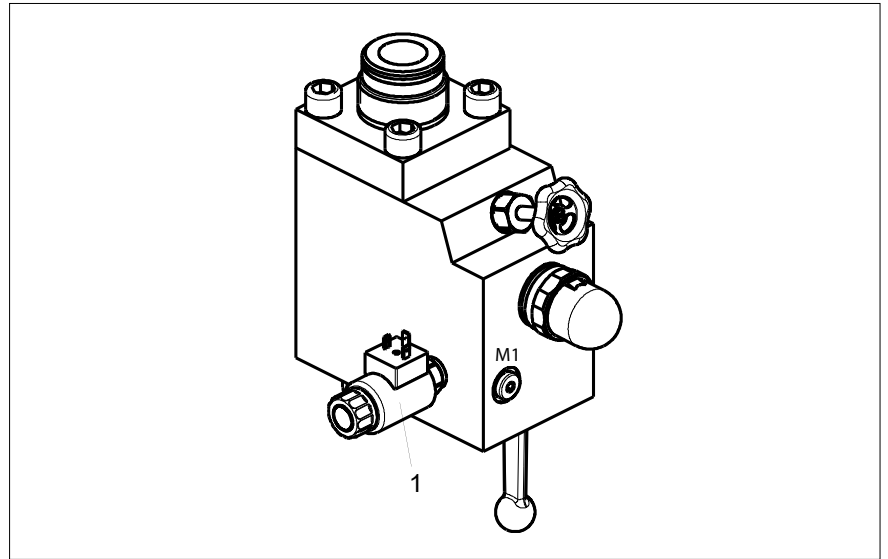


Fig. 10: Electrical connections ABZSS... E

1 Directional seat valve

Directional seat valve (1) Direct operated directional seat valve for electro-magnetic unloading.

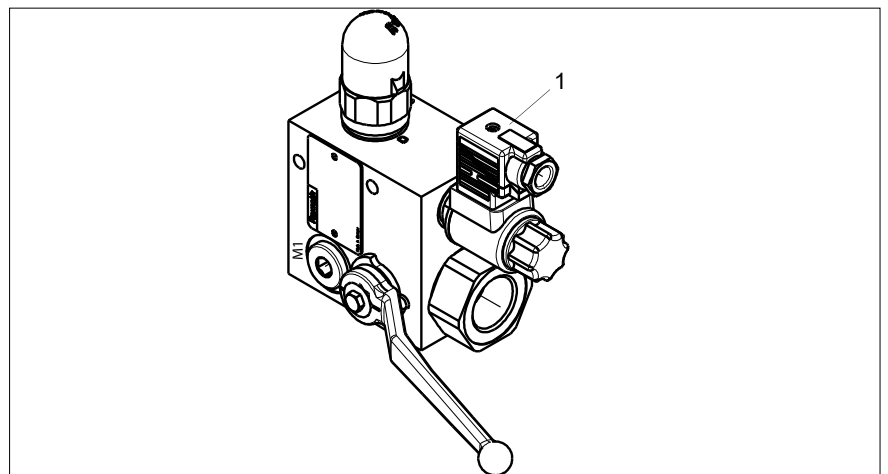


Fig. 11: Accumulator shut-off block 0532VAW/4

1 Directional seat valve

Directional seat valve (1) Direct operated directional seat valve for electro-magnetic unloading.

Product description

4.4 Sizes

The functions of the accumulator shut-off blocks size 08, 10, 20 and 30 are identical. The designs are only different with regard to their size, weight and performance data.

Edge dimensions of the blocks

Table 3: Edge dimensions of the manifolds

Size	Size / mm
ABZSS08	45 x 95 x 110
ABZSS10	67 x 110 x 100
ABZS20	75 x 130 x 120
ABZS30	103 x 175 x 160
ABZSS30...SO30	103 x 200 x 182
ABZSS-P30....	155 x 175 x 190
0532VAW20	63 x 100 x 100
0532VAW30	120 x 120 x 137

Weight indications see chap. 5.1.1

4.5 Operating elements

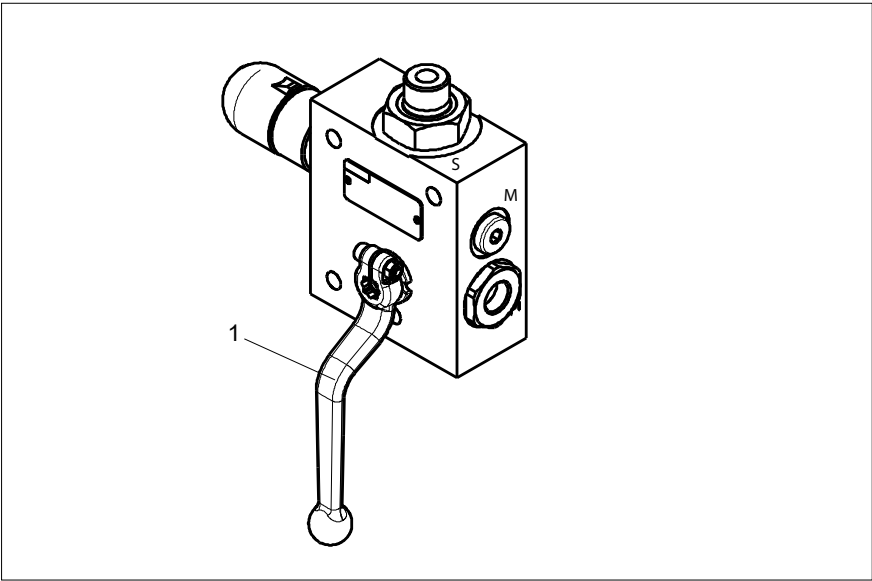


Fig. 12: Operating elements Type ABZSS08M

- 1 System shut-off cock with manual unloading of the pressure

System shut-off cock with manual unloading of the pressure (1)

System shut-off cock with manual unloading of the pressure

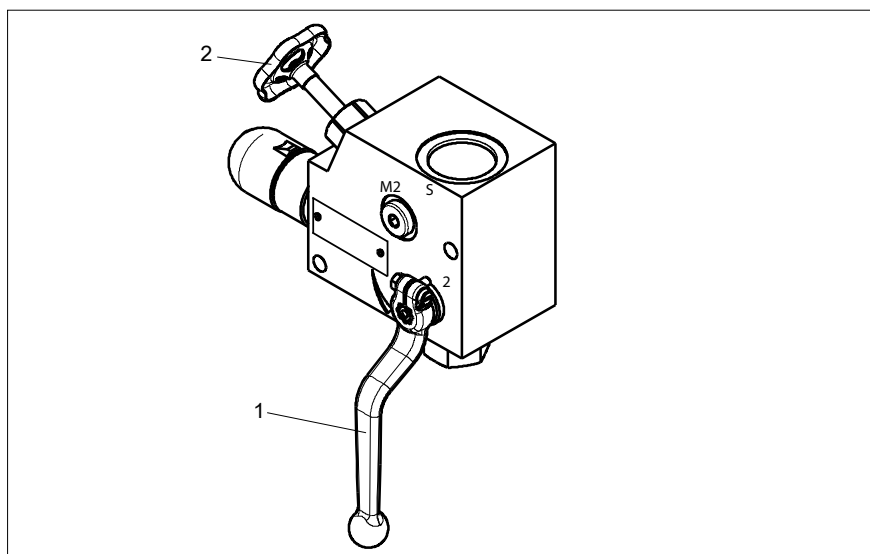


Fig. 13: Operating elements ABZSS10M, ABZSS10M and ABZSS30M

1 System shut-off cock

2 Manual unloading

System shut-off cock (1)

System shut-off cock serves the isolation of the pressure line

Manual unloading (2)

Manual unloading, serves the manual unloading of the accumulator pressure

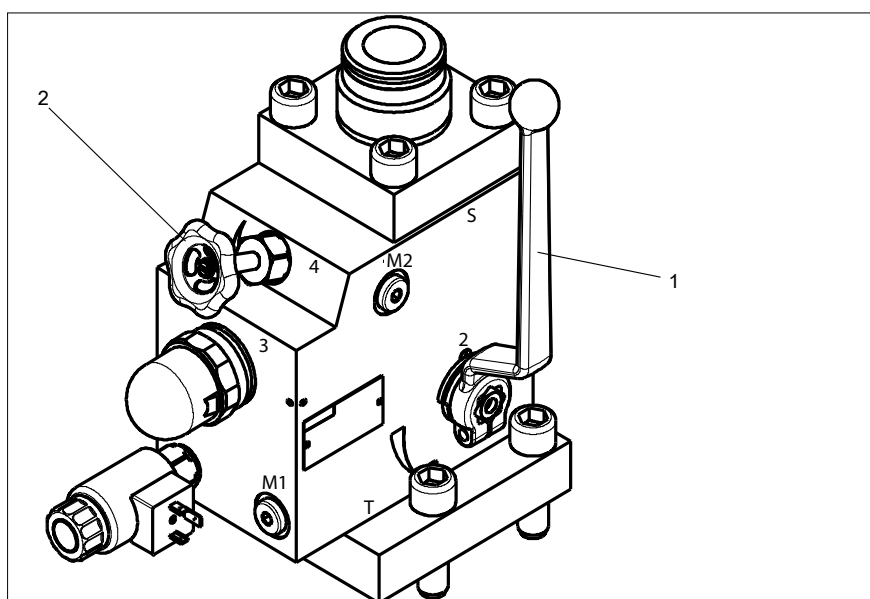


Fig. 14: Operating elements ABZSS-P30M

1 System shut-off cock

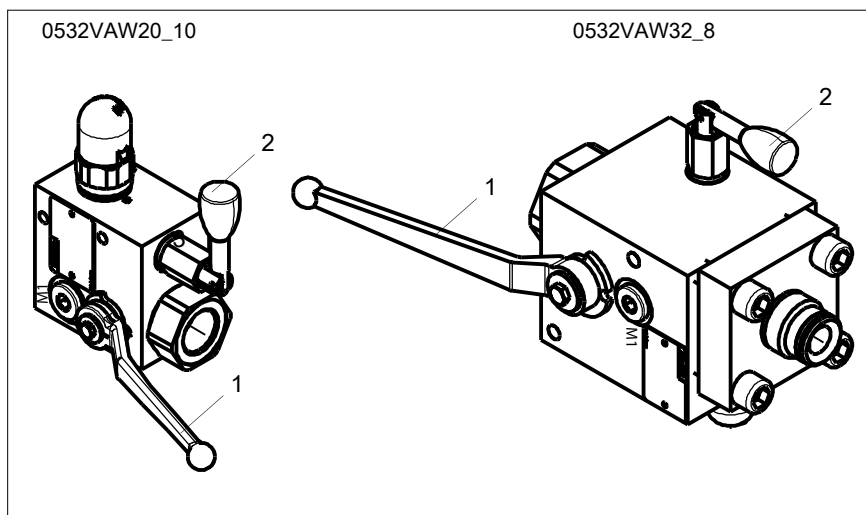
2 Manual unloading

System shut-off cock (1)

System shut-off cock serves the isolation of the pressure line

Manual unloading (2)

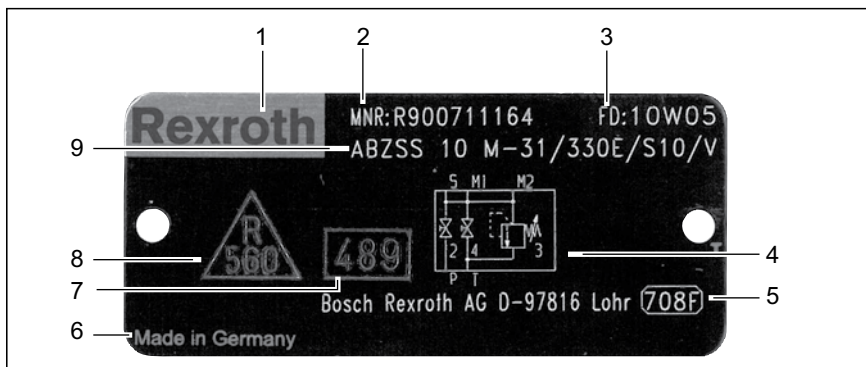
Manual unloading, serves the manual unloading of the accumulator pressure

Product description**Fig. 15: Operating element 0532VAW20_10 and 0532VAW32_8**

- | | |
|---|--|
| <p>System shut-off cock (1)</p> <p>Manual unloading of the pressure (2)</p> | <p>1 System shut-off cock</p> <p>System shut-off cock serves the isolation of the pressure line</p> <p>2 Manual unloading</p> <p>Serves the manual unloading of the accumulator pressure</p> |
|---|--|

4.6 Manifold identification

The manifold can be identified using its name plate. The following figure shows an exemplary presentation.

**Fig. 16: Example name plate manifold ABZSS10M**

- | | |
|---|---|
| <p>1 Manufacturer</p> <p>2 Material number</p> <p>3 Date of manufacture</p> <p>4 Circuit diagram symbol</p> <p>5 Area / works number</p> | <p>6 Designation of origin</p> <p>7 Personal stamp of the fitter</p> <p>8 Personal stamp of the examiner</p> <p>9 Material short text</p> |
|---|---|

Product description

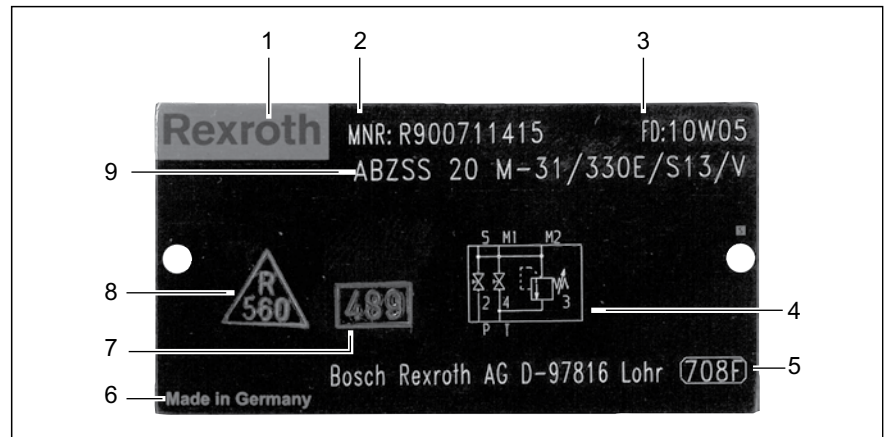


Fig. 17: Example name plate manifold ABZSS20M

- | | |
|--------------------------|----------------------------------|
| 1 Manufacturer | 6 Designation of origin |
| 2 Material number | 7 Personal stamp of the fitter |
| 3 Date of manufacture | 8 Personal stamp of the examiner |
| 4 Circuit diagram symbol | 9 Material short text |
| 5 Area / works number | |

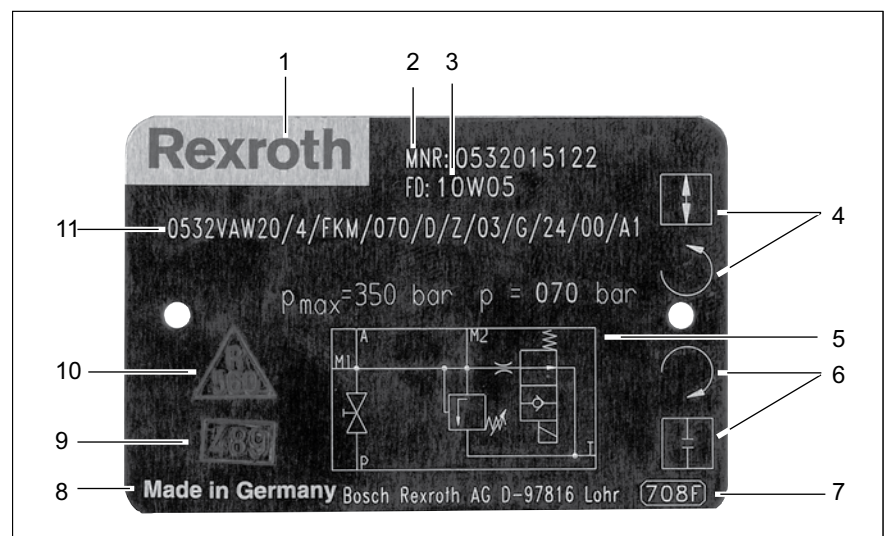


Fig. 18: Example name plate manifold 0532VAW20

- | | |
|--|-----------------------------------|
| 1 Manufacturer | 7 Area / works number |
| 2 Material number | 8 Designation of origin |
| 3 Date of manufacture | 9 Personal stamp of the fitter |
| 4 Direction of rotation "open" system shut-off cock | 10 Personal stamp of the examiner |
| 5 Circuit diagram symbol | 11 Material short text |
| 6 Direction of rotation "close" system shut-off cock | |

5 Transport and storage

5.1 Transporting the manifold

5.1.1 Manual transport

CAUTION! 	Risk of injury to persons undertaking manual lifting/handling due to overloading and incorrect posture! In the case of lifting/handling, there is the risk of damage to health due to overloading and incorrect posture. ▶ The limits of the respective persons involved in manual lifting/handling must be realistically assessed. Muscular strength and constitution differs considerably among human beings. ▶ Wherever possible, suitable lifting aids should be used such as e.g. carrying straps. ▶ Always use appropriate techniques when lifting, setting down and moving products.
CAUTION! 	Risk of damage! Impacts or similar forces may damage add-on units of the manifold. ▶ Do not expose the add-on units to mechanical loads under any circumstances. ▶ Never use the product as a handle or step. ▶ Do not place any objects on top of the manifold. ▶ Be very careful when transporting the manifold.

5.1.2 Transport by means of lifting straps and attachment points

DANGER! 	Risk of personal injury and damage to property from parts falling down! Manifolds may fall down during transport and lead to fatal injuries. ▶ Put the packaging on a level bearing surface.
---	---

The manifold ABZSS30... and ABZSS-P30... can be transported using a chain hoist.

- ▶ Make sure that the chain hoist's lifting capacity is sufficiently dimensioned in order to safely bear the manifold's weight.

Weight **Table 4: Weight manifolds**

Type	Weight / kg (ca.)
ABZSS08	4.0
ABZSS10	5.5
ABZSS20	8.8
ABZSS30	20.8
ABZSS30...SO30	26.8
ABZSS-P30	26.0
0532VAW20	12.3
0532VAW 30	33.4

Transport and storage**Tapped hole for adjusting screw**

For the safe transport of the manifold ABZSS30... and ABZSS-P30..., an adjusting screw can be screwed into the tapped holes for the accumulator adapter.

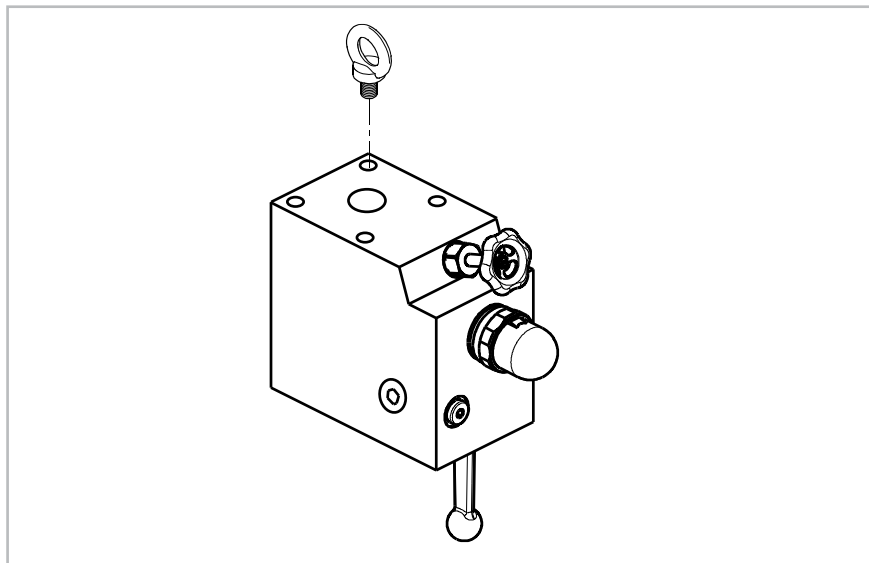


Fig. 19: Tapped holes for adjusting screws (eyebolts)

Transporting the manifold

- ▶ Screw the adjusting screw (eyebolt) into the tapped hole until it is hand-tight.
- ▶ Fasten a belt or chain at the adjusting screw.
- ▶ Hang the belt or chain to a crane hook.
- ▶ Make sure that the belt or chain cannot damage add-on units during lifting.
- ▶ Lift the manifold slowly and carefully in order to avoid oscillations of the manifold.
- ▶ Transport the manifold to the desired location.

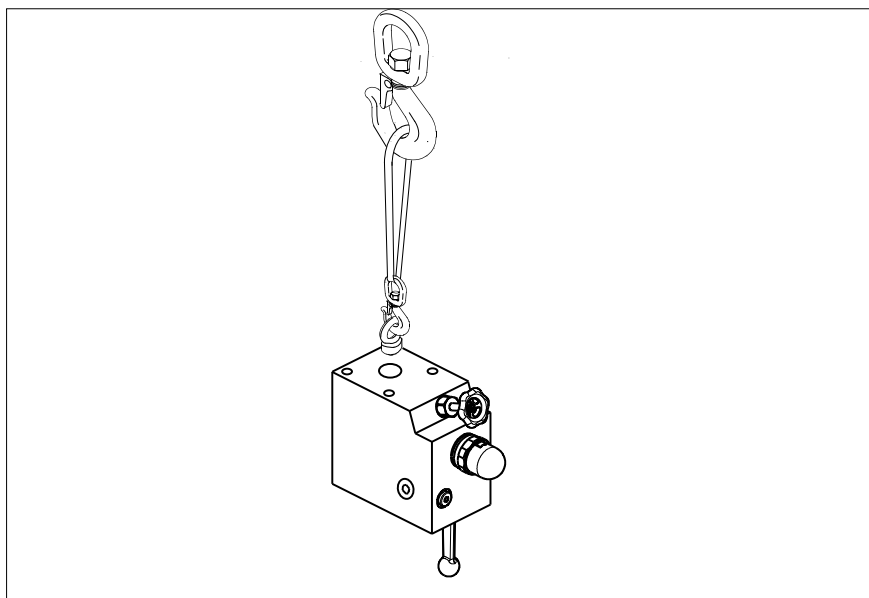


Fig. 20: Example: transporting the manifold using lifting tools adjusting screw (eyebolt)

Transport and storage

5.2 Storing the manifold

For storing the manifold, you have to comply with the following environmental conditions:

Storage conditions

Table 5: Storage conditions

Denomination	Area
Temperature range	10 - 25 °C
Relative humidity (no condensation)	max. 65 %
UV protection	100 %
Condensation	None
Additional ozone formation close to the storage	None

Storage up to 3 months

No special preservation necessary.

Storage longer than 3 months

Contact Bosch Rexroth for the preservation and later commissioning of the manifold if the manifold must be stored for a period of more than three months.

6 Assembly

6.1 Unpacking

CAUTION!



Danger of parts that are falling down!

If their packing is opened improperly, manifolds may roll out and cause injuries or damage.

- ▶ Put the packaging on a level bearing surface.

- ▶ Remove the product packaging.

- ▶ Dispose of the packaging in accordance with the currently applicable national regulations in your country.

6.2 Installation conditions

- ▶ For installing the product always observe the environmental conditions specified in chapter 15 "Technical data".

Fastening

- ▶ Due to their own weight and the additional reaction forces generated in the hydraulic accumulator due to the hydraulic fluid, the manifold and the attached hydraulic accumulators must be sufficiently secured and attached. The hydraulic accumulators are to be fastened so that in case of operational vibrations, safe hold is guaranteed. The accumulator manufacturers offer corresponding retaining clamps and consoles as accessory. Also observe the documentation of the accumulator manufacturer.

Cleanness

- ▶ It is imperative to provide for absolute cleanness. The manifold and all other parts used must be installed free from any dirt. Contamination of the hydraulic fluid may impair the service life as well as the function of the manifold considerably.

Line connections to the manifold

- ▶ In case of hydraulic hose connections at inlets or outlets of the manifold, e.g. for connecting components, the relevant standards and country-specific rules are to be observed. When designing the hose connection, you have to assume the possible hydraulic and mechanical maximum loads considering the necessary safety allowances that may crop up in the different operating conditions. The latter have to be determined separately, like e.g. positive or negative pressure peaks. The operating and maintenance instructions of the components and hose connections must also be observed.

Fastening screws

- ▶ For the assembly of the manifold, use fastening screws according to the dimensions specified in the catalog.
- ▶ The tightening torque and the property class of the fastening screws are to be observed.

Pressure relief valve

The pressure relief valve cannot be exchanged without destroying the sealing between pressure relief valve and manifold.

Assembly

6.3 Assembling the manifold

DANGER!**Risk of personal injuries and damage to property!**

Assembly of the manifold requires basic mechanical and hydraulic knowledge.

- ▶ The manifold may only be assembled by qualified personnel (see chapter 2.3 "Qualification of personnel").

6.3.1 Assembling the manifold at the accumulator

DANGER!**Risk of injuries due to insufficient mounting of the manifold!**

An insufficiently mounted manifold may result in uncontrolled movements, damage other system components and fall down.

- ▶ Make sure that the accumulator is fastened according to the specifications of the accumulator/machine manufacturer.

CAUTION!**Risk of damage!**

Impacts or similar forces may damage add-on units of the manifold.

- ▶ Do not expose the add-on units to mechanical loads under any circumstances.
- ▶ Attach chains or belts so that the add-on units are not damaged during transport.
- ▶ Never use the product as a handle or step.
- ▶ Do not place any objects on top of the manifold.
- ▶ Be very careful when assembling the manifold.

For assembling the manifold, the following work steps are required:

- ▶ Get the parts list and the installation plan of the system as well as the operating instructions of the manifold ready.
- ▶ Remove the protective plugs from the manifold.
- ▶ Check all block contact surfaces and accumulators for cleanliness and remove protective plugs, if necessary.
- ▶ Lift the manifold using suitable lifting tools. In this connection, observe the information in chapter 5.1 "Transporting the manifold".
- ▶ Make sure the relevant system component is not under pressure or voltage before assembling the product or when connecting and disconnecting plug-in connectors.
- ▶ Protect the system against unintended re-start during the assembly.
- ▶ Lay cables and lines so that they cannot be damaged and no one can trip over them.
- ▶ When assembling, provide for absolute cleanness in order to prevent contamination from getting into the hydraulic piping and causing product wear or malfunctions.

- ▶ In case of hydraulic hose connections at inlets or outlets of the manifold, e.g. for connecting components, the relevant standards and country-specific rules are to be observed. When designing the hose connection, you have to assume the possible hydraulic and mechanical maximum loads considering the necessary safety allowances that may crop up in the different operating conditions. The latter have to be determined separately, like e.g. over-/under-pressure peaks. The operating and maintenance instructions of the components and hose connections must also be observed.
- ▶ Check the hydraulic lines for cleanness.
- ▶ If necessary, secure seal rings against slipping using installation grease.
- ▶ Before commissioning, make sure that all the seals and plugs of the plug-in connections are installed correctly and undamaged to ensure that they are leakproof and fluids and contamination parts are prevented from penetrating the product.
- ▶ In case of chip-removing, grinding or welding works close to our delivered hydraulic products, protective measures against contamination and/or subsequent extensive cleaning of the components concerned are to be ensured.

6.3.2 Assembling hydraulic accumulators at the manifold

For assembling the accumulators at the manifold, the following work steps are required:

- ▶ Get the assembly drawing and the parts list of the manifold, the installation plan of the system as well as the operating instructions of the accumulator manufacturer ready.
- ▶ Lift the accumulators using suitable lifting tools. In this connection, observe the specifications of the accumulator manufacturer.
- ▶ Assemble the pressure accumulator according to the assembly manufacturer. In this connection, observe the assembly information of the flange and accumulator manufacturers.
- ▶ Moreover secure the pressure accumulators with clamps against vibrations. In this connection, observe the specifications of the machine and accumulator manufacturer.

6.3.3 Hydraulically connecting the manifold

CAUTION!



Risk of injury when assembling under pressure!

If you fail to depressurize the product before starting the assembly, you may suffer injury and also damage the unit or system components.

- ▶ Always depressurize the relevant system part before assembling the product.
 - ▶ Protect the system against unintended re-start during the assembly.
-

Assembly

WARNING!**Risk of personal injuries and damage to property!**

If the operating pressure of the line connections (hose and/or pipe) at the inlets or outlets of the manifold is lower than that of the type-tested pressure relief valve, persons may be killed by bursting line connections.

- ▶ Relevant standards and country-specific regulations must be observed. When designing the line connections (hose and/or pipe), you have to assume the possible hydraulic and mechanical maximum loads considering the necessary safety allowances that may crop up in the different operating conditions. The latter have to be determined separately, like e.g. positive and negative pressure peaks. The operating and maintenance instructions of the components must be observed.

WARNING!**Risk of personal injuries and damage to property!**

If due to the increasing flow, the system pressure increases, persons may be killed by bursting line connections and/or components.

- ▶ According to PED 97/23/EC, the increase in system pressure caused by the flow must not exceed 10 % of the set response pressure.
- ▶ The discharge line (port T) of the manifold must always end in a non-dangerous manner. In the discharge system, the accumulation of fluids must not be possible (citation AD2000 data sheet A2).
- ▶ The discharge line (port T) must be dimensioned so that the maximum admissible flow is considered depending on the backpressure in the discharge line according to the technical data. See technical data sheet of the manifold ABZSS data sheet RE 50121 and of the manifold 0532VAW data sheet RE 50128.

CAUTION!**Risk of personal injuries and damage to property!**

If the hydraulic piping is not laid in a professional way, you may suffer injury or damage the hydraulic piping.

- ▶ Lay the hydraulic piping in a professional way, so that you cannot trip over the hydraulic piping.

CAUTION!**Damage to the manifold!**

When assembling hydraulic piping and hoses under mechanical stress, they are exposed to additional mechanical forces during operation which reduce the service life of the connection elements at the manifold.

- ▶ Assemble lines and hoses without mechanical stress.

CAUTION!**Wear, tear and malfunctions!**

The cleanliness of the hydraulic fluid has a considerable impact on the cleanliness and service life of the hydraulic system as a whole. Any pollution/contamination of the hydraulic fluid will result in wear and malfunctions. In particular, contamination parts like e.g. welding beats or metal cuttings in the hydraulic piping may damage the manifold.

- ▶ Always ensure absolute cleanliness.
- ▶ Assemble the manifold free from any pollution.
- ▶ Make sure that all connections, hydraulic piping and add-on units (e. g. measuring instruments) are clean.

- ▶ In case of chip-removing, grinding or welding works close to our delivered hydraulic products, protective measures against contamination and/or subsequent extensive cleaning of the components concerned are to be ensured.
- ▶ Ensure that no pollutants are able to penetrate when sealing the connections.
- ▶ Ensure that no detergents are able to penetrate the hydraulic system.
- ▶ Do not use cleaning rags/cotton waste or linty cloths for cleaning.

For the hydraulic connection of the manifold, the following work steps are required:

- ▶ Familiarize with the position of the hydraulic connections. In this connection, observe chapter 4.3 "Interfaces".
- ▶ Check the hydraulic connections for cleanliness and remove protective plugs, if necessary.
- ▶ Check the hydraulic lines for cleanness.
- ▶ Make sure that the line connections contain the intended seals.
- ▶ If necessary, secure O-rings against slipping using installation grease.
- ▶ Connect the lines "P" and "T" according to the specifications of the system or machine manufacturer. Observe the tightening torques of the mounting manufacturer.

The hydraulic supply is connected.

6.3.4 Connecting to the electric supply

CAUTION!



Risk of injury when assembling under voltage!

If you do not switch off the voltage before assembling the product, you may get injured or the device or system components may be damaged.

- ▶ Always switch off power supply to the relevant system component before assembling the product.

CAUTION!



Connecting or disconnecting plug-in connectors under voltage damages electric components!

Connecting or disconnecting plug-in connectors under voltage may destroy the electric components at the manifold.

- ▶ Always switch off power supply to the relevant system component before connecting or disconnecting plug-in connectors.

CAUTION!



Risk of personal injuries and damage to property!

If the electric line is not laid in a professional way, you may suffer injury or damage the electric line.

- ▶ Lay the electric line in a professional way, so that you cannot trip over the electric line.

Assembly

For the electrical connection of the manifold, the following work steps are required:

- ▶ Familiarize with the position of the electrical connections. In this connection, observe chapter 4.3 "Interfaces".
- ▶ Regarding the electric wiring at the block, observe the data sheets of the hydraulic components available for this purpose.
- ▶ Check the electrical connections for cleanness.
- ▶ Now connect the manifold electrically according to the specifications of the system or machine manufacturer.

The electric supply is connected.

6.4 Assembly of the pressure relief valve

For assembling the pressure relief valve, the following work steps are required:

- ▶ Remove the protective cap from the pressure relief valve. Remove the single seal ring or both seal rings contained in the protective cap.
- ▶ If the pressure relief valve comprises two seal rings, clamp the smaller seal ring into the larger seal ring.
- ▶ Insert the seal ring or the seal ring combination into the mounting cavity, ensure the most concentric position in the mounting cavity possible and contact of the whole surface.
- ▶ Screw in the pressure relief valve and tighten it applying the tightening torque specified in table 6.

Table 6: Tightening torque depending on manifold type, pressure relief valve and size

Manifold type ABZSS				
Valve type and size ¹⁾				
Part marking				
Pressure rating	DBD6 849	DBD10 850, 390	DBD20 361	DBD30 362
up to 210 bar	50 ± 5 Nm	100 ± 5 Nm	150 ± 10 Nm	350 ± 20 Nm
up to 350 bar	80 ± 5 Nm	150 ± 5 Nm	300 ± 10 Nm	500 ± 20 Nm

Manifold type 0532VAW	
Pressure rating	
up to 330 bar	70 + 5 Nm ¹⁾

¹⁾ Oiled screws, tightening using a torque-controlled tool of accuracy class ±10 %.

7 Commissioning

Hydraulic products are generally intended for installation in machines/systems or devices. The function of the hydraulic product must thus always be seen in connection with the function of this machine - i.e. externally identical hydraulic products have - due to the function of the machine into which they are installed - a different functional behavior.

CAUTION!**Damage to the manifold!**

Commissioning of the manifold requires basic mechanical and hydraulic knowledge.

- Only qualified personnel (see section 2.3 "Qualification of personnel") is authorized to commission the manifold.
-

7.1 First commissioning

CAUTION!**Risk of damage to property!**

Polluted hydraulic fluid could result in wear and malfunctions. In particular, contamination parts like e.g. welding beads or metal cuttings in the suction lines may damage the manifold.

- When commissioning you should ensure absolute cleanliness.
 - Ensure that no pollutants are able to penetrate when sealing the measuring ports.
-

7.1.1 Preparing the commissioning

CAUTION!**Risk of personal injuries and damage to property!**

If the manifold is not mounted correctly, persons could be injured or the product or system could be damaged when commissioning the manifold.

- Make sure that the manifold has been mounted correctly by qualified personnel before commissioning the manifold.
-

DANGER!**Risk of personal injuries and damage to property!**

If the pressure adjustment of the type tested pressure relief valve has a higher setting pressure than the maximum operating pressure of the hydraulic accumulator, the hydraulic accumulator may burst and consequently, persons may be killed.

- The pressure adjustment of the type tested pressure relief valve must not be higher than the maximum operating pressure of the hydraulic accumulator. Compare the technical data of the hydraulic accumulator with that of the manifold.
-

Commissioning

WARNING!**Risk of personal injuries and damage to property!**

Leaking oil mist due to defective or improperly assembled seals may cause allergic reactions, fire and/or explosions as well as environmental contamination.

- ▶ Open sources of ignition and welding works at the machine are only admissible with depressurized hydraulic systems.

WARNING!**Slip hazard!**

Due to oily surfaces as a consequence of maintenance measures, there is a slip hazard around the manifold.

- ▶ Always move very carefully.
- ▶ Take suitable precautions.

Observe the specifications regarding the oil cleanliness in chapter 15 "Technical data". Proceed as follows to commission the manifold:

- ▶ Pre-load the hydraulic accumulators at the manifold according to the specifications of the machine manufacturer. In this connection, observe the information in the operating instructions of the hydraulic accumulators.
- ▶ Make sure that the gas pre-load of the accumulators complies with the technical specifications of the machine manufacturer.
- ▶ Until the complete bleeding, commission the system with reduced pressure.
- ▶ Make sure that all connections are either used or covered.
- ▶ Check the hydraulic lines for tightness.
- ▶ Only commission hydraulic drives after you have on the one hand extensively familiarized with the function of the hydraulic product and the hydraulic devices used and on the other hand with the hydraulically driven machine functions and clarified and remedied possible risks.
- ▶ Commission the product only if it is installed completely.
- ▶ Leaking oil mist due to defective or improperly assembled seals may cause allergic reactions, fire and/or explosions as well as environmental contamination. Open sources of ignition and welding works at the machine are only admissible with depressurized hydraulic systems.
- ▶ If leakage is determined at pressurized components, immediate shut-off and cause analysis are required in order to be able to initiate corrective measures, if possible.

7.1.2 Performing the commissioning

For commissioning, the following work steps are required:

- ▶ Get the data sheets of the components to be assembled according to the parts list ready.
- ▶ Keep the suitable tool ready.
- ▶ Compare the pressure adjustment of the type tested pressure relief valve with the maximum admissible operating pressure of the accumulator. The pressure adjustment of the type tested pressure relief valve must not be higher than the admissible operating pressure of the accumulator.
- ▶ Close the manual unloading.
- ▶ Open the system shut-off cock.

Bleeding

- ▶ On the actuator side, all channels are to be bled. In this connection, observe the specifications of the machine manufacturer.

Checking the 2-way cartridge valve

- ▶ Check the electric signals. (optional)

Commissioning of the manifold is completed.



Sealed safety valves must never be adjusted!

7.2 Re-commissioning after extended standstill

3-month standstill

Contact Bosch Rexroth for the re-commissioning after a standstill of more than 3 months.

Re-commissioning

In order to re-commission the manifold, proceed as described in chapter 7.1 "First commissioning".

Operation

8 Operation

The manifold may only be operated with the data admissible according to chapter 15 "Technical data".

The manifold may only be operated if it is in an unobjectionable condition.

Only authorized specialized staff is allowed to operate the adjustment devices of the components or parts, under the proviso that the hydraulic system is used as intended.

CAUTION!**Risk of burning!**

The manifold and the on/off valve solenoids may become hot during operation and cause burns in case of contact.

- ▶ Wear personal protective equipment.

WARNING!**Risk of personal injuries and damage to property!**

Leaking oil mist due to defective or improperly assembled seals may cause allergic reactions, fire and/or explosions as well as environmental contamination.

- ▶ Immediately shut-off the system and carry out a cause analysis.
- ▶ Open sources of ignition and welding works at the machine are only admissible with depressurized hydraulic systems.

WARNING!**▶ Slip hazard!**

Due to oily surfaces as a consequence of maintenance measures, there is a slip hazard around the manifold.

- ▶ Always move very carefully.
- ▶ Take suitable precautions.

In order to guarantee long and reliable functioning of the manifold, Bosch Rexroth recommends checking the hydraulic system and the manifold regularly:

- Continuously monitor noises, vibrations and temperatures.
- After some operating time, check the hydraulic fluid in the tank for bubble or foam formation at the surface.

Changes in operating speeds, temperatures, noise increase or power consumption are an indication of wear or damage at the system or the manifold.

9 Maintenance and repair

DANGER!



Risk of personal injuries and damage to property!

The maintenance and repair of the manifold requires basic mechanical and hydraulic knowledge.

- The manifold may only be maintained and repaired by qualified personnel (see chapter 2.3 "Qualification of personnel").

9.1 Cleaning and care (maintenance)

CAUTION!



Penetrating dirt and liquids will cause faults!

Safe function of the manifold is no longer ensured.

- When commissioning you should ensure absolute cleanliness.
- Do not use a high-pressure cleaner.

CAUTION!



Damage to the surface from solvents and aggressive detergents!

Aggressive detergents may damage the seals of the manifold and let them age faster.

- Never use solvents or aggressive detergents.
- Do not use a high-pressure cleaner for cleaning.

- Cover all openings with appropriate protective caps.
- Check that all seals and plugs of the plug-in connections are firmly fitted so that no humidity can penetrate the manifold during cleaning.

9.2 Maintenance plan

Prescribed and extraordinary maintenance works are to be documented in a meaningful way and stored for evidence purposes. If required maintenance works cannot be proven, the warranty may forfeit as, for example, excessive wear may occur and intended operating times may not be reached.

Machine manufacturer or operator

Table 7: Maintenance plan operator or machine manufacturer

Measures	Intervals	Comments
Visual inspection for leakage at the manifold	Specification machine manufacturer and operator	
Fluid samples or analyses	After commissioning	Fresh oil sample, observe safety data sheet, document product name incl. charge number
	50 operating hours after commissioning	
	4 weeks after commissioning	
	12 weeks after commissioning	
	Every six months	

Maintenance and repair

9.3 Maintenance

Perform the prescribed maintenance works in the intervals specified in the operating instructions and in the instruction manual of the machine manufacturer.

- ▶ In maintenance or repair works, only genuine spare parts or parts approved of by Rexroth may be used. If non-original parts are used, any liability or warranty claims that might arise in connection with these parts against Bosch Rexroth are forfeited.
- ▶ Make sure that no line connections, ports and components are disconnected as long as the system is under pressure and voltage or the hydraulic accumulators are still under hydraulic pressure. For an additional manual unloading of the pressure, the manual unloading at the manifold has to be opened before any maintenance and servicing works. Protect the system against unintended re-start.
- ▶ After completion of the repair works make sure that the fluid connection is connected as provided again and the gas connection is tightened and covered.

DANGER!

Risk of personal injuries and damage to property!

Maintenance of the manifold requires basic mechanical and hydraulic knowledge.

- ▶ The manifold may only be maintained by qualified personnel (see chapter 2.3 "Qualification of personnel").

CAUTION!

Danger of burns at hot surfaces!

Manifold and on/off valve solenoid may become hot during operation and cause burns in case of contact.

- ▶ Wear personal protective equipment.

WARNING!

Risk of personal injuries and damage to property!

Leaking oil mist due to defective or improperly assembled seals may cause allergic reactions, fire and/or explosions as well as environmental contamination.

- ▶ Open sources of ignition and welding works at the machine are only admissible with depressurized hydraulic systems.

WARNING!

Danger of burns due to hot hydraulic fluids!

Hot hydraulic fluid may leak when loosening components or connections and cause serious injuries.

- ▶ Before loosening components or connections wait until the hydraulic fluid has cooled down or take suitable precautions.

WARNING!

Slip hazard!

Due to oily surfaces as a consequence of maintenance measures, there is a slip hazard around the manifold.

- ▶ Always move very carefully.
- ▶ Take suitable precautions.

Preparing maintenance works

Before starting the maintenance works, perform the following work steps:

- ▶ Get the data sheets of the components to be assembled and disassembled according to the parts list ready.
- ▶ Get a suitable, clean tank ready for temporarily storing the valve to be disassembled.
- ▶ Keep suitable tools ready.
- ▶ Open the manual unloading.
- ▶ Close the system shut-off cock.

**9.4 Repair**

Bosch Rexroth offers a wide range of repair services for the manifold.

- ▶ Only use genuine spare parts from Bosch Rexroth for repairing the Rexroth product.
- ▶ Tested and pre-assembled original Rexroth assemblies allow for successful repair requiring only little time.

9.5 Spare parts**CAUTION!****Damage to property and personal injuries due to faulty spare parts!**

Spare parts that do not meet the technical requirements specified by Bosch Rexroth may cause personal injuries and damage to property.

- ▶ Only use genuine spare parts from Bosch Rexroth.

- ▶ Order spare parts in writing.
- ▶ Please send your spare parts order to the Bosch Rexroth service next to you or directly to the headquarters (see "Address directory" in chapter 16 "Appendix").
- ▶ When ordering spare parts, please indicate the following information:
 - The material number of the required spare part
 - The Bosch Rexroth order number (SAP commissions No.)
 - The order date
- ▶ Additionally indicate:
 - The desired number of spare parts
 - The desired type of dispatch (e. g. as parcel, freight, air freight, by courier service, etc.).

Please refer to the address directory in chapter 16 "Appendix" for the addresses of our foreign subsidiaries.

10 Decommissioning

The manifold is a component that does not require decommissioning. As a result, this chapter does not contain any information. For details about how to disassemble or replace your manifold, please refer to chapter 11 "Disassembly and replacement".

11 Disassembly and replacement

11.1 Required tools

Observe the specifications of the machine manufacturer.

You moreover need:

- Suitable tools for opening the screw connections
- Suitable lifting gear
- Oil tray

11.2 Preparing for disassembly

Prepare the disassembly of the manifold as follows:

1. Decommission the entire system as described in the overall system manual.
2. Discharge the pressure of the hydraulic accumulators via the manual unloading.
3. Keep the disassembly tools ready.

You can now start the disassembly.

11.3 Disassembling the manifold

DANGER!**Risk of personal injuries and damage to property!**

Disassembly of the manifold requires basic mechanical and hydraulic knowledge.

- The manifold may only be disassembled by qualified personnel (see chapter 2.3 "Qualification of personnel").

CAUTION!**Danger of burns at hot surfaces!**

Manifold and on/off valve solenoid may become hot during operation and cause burns in case of contact.

- Wear personal protective equipment.
-

WARNING!**Danger of burns due to hot hydraulic fluids!**

Hot hydraulic fluid may leak when loosening components or connections and cause serious injuries.

- Before loosening components or connections wait until the hydraulic fluid has cooled down or take suitable precautions.

DANGER!**Risk of injuries when disassembling under pressure!**

If you do not discharge the pressure before disassembling the product, you may get injured or the device or system components may be damaged.

- Always depressurize the relevant system part before disassembling the product.

Proceed as follows to disassemble the manifold:

1. Make sure that the whole system is depressurized.
2. Open the manual unloading at the manifold.
3. Make sure that the accumulator is depressurized.
4. Disassemble the hydraulic piping.
5. Disassemble the hydraulic accumulators.
6. Disassemble the manifold according to the specifications of the machine manufacturer.

The manifold is disassembled.

11.4 Preparing the manifold for storage/further use

For the storage, observe the storage provisions in chapter 5.2 "Storing the manifold".

11.5 Replacing the manifold

- Disassemble the manifold as described in chapter 11 "Disassembly and replacement".
- Assemble the manifold as described in chapter 6 "Assembly".

12 Disposal

12.1 Environmental protection

Irresponsible disposal of the manifold may result in environmental pollution.

- ▶ Dispose of the manifold in accordance with the currently applicable national regulations in your country.
- ▶ Disassemble the manifold into its individual components in order to recycle them.
- ▶ Dispose of hydraulic fluid residues according to the respective safety data sheets valid for these hydraulic fluids.
- ▶ Separate:
 - Cast iron
 - Steel
 - Non-ferrous metal
 - Seals
 - Plastic
 - Electronic components

13 Extension and conversion

For the extension or conversion of the manifold, a written release by Bosch Rexroth is necessary.



The Bosch Rexroth warranty applies only to the delivered configuration and extensions considered in this configuration. After an extension or a conversion without written release by Bosch Rexroth, the warranty will forfeit.

14 Troubleshooting

14.1 How to proceed for troubleshooting

- Always work systematically and focused, even when under time pressure. Random and imprudent disassembly and readjustment of settings can, in the worst-case scenario, result in the inability to determine the original cause of the fault.
- First get a general idea of how your product works in conjunction with the entire system.
- Try to find out whether the product has worked properly in conjunction with the entire system before the error occurred first.
- Try to determine any changes of the entire system in which the product is integrated:
 - Were there any changes to the product's operating conditions or operating range?
 - Have modifications (e. g. refittings) or repairs been carried out at the overall system (machine/system, electrical systems, control) or at the product?
If so: What were they?
 - Was the product or machine used as intended?
 - How did the malfunction become apparent?
- Try to get a clear idea of the cause of the fault. Ask the direct (machine) operator, if necessary.

Troubleshooting

Table 8: Malfunction list manifold

Fault	Possible causes	Remedy
Accumulator overloaded	Unsuitable hydraulic fluid	Admissible hydraulic fluid, see data sheet
	Viscosity too high	Admissible hydraulic fluid, see data sheet
	Insufficient oil cleanliness	Cleanliness class, see data sheet
	Pressurized tank line	Lead back tank line into the tank in a pressureless form
	Maximum, securable flow too little	Maximum admissible flow, characteristics see data sheet
	Pressure relief valve does not switch	Check usability of the type tested pressure relief valve
Supply flow to the accumulator too little	Oil supply	Switch on pump
	Leakage to the tank	Selection of the pressure relief valve according to data sheet
	Manual unloading not completely closed	Close manual unloading
	In closed condition, the manual unloading is not tight	Use an original manual unloading
	Electro-magnetic unloading open	Control the electro-magnetic unloading
	In closed condition, the electro-magnetic unloading is not tight	Use an original electro-magnetic unloading
	External leakage	Use original seals
Accumulator damaged, no pressure accumulation	Discharge oil insufficiently limited	Accumulator unloading velocity limited by orifice
External leakage	Housing not tight	Use original seals
	Accumulator adapter not tight	Use original seals
	Sealing surfaces contaminated or damaged	Ensure cleanness and integrity of the sealing surfaces
	Incorrect assembly	Assembly only by authorized, trained and instructed specialized staff; only use original spare parts
Electro-magnetic unloading without function	Electrical voltage supply	Check the voltage supply of the machine manufacturer
Loud flow noises	Air in the system	Bleed the system, contact the machine manufacturer
	Manual unloading not completely closed	Close manual unloading
Displacement cannot be completely isolated	Leak at the shut-off cock	Use original shut-off cock

15 Technical data

Standards and directives

Table 9: Standards and directives used

Standards and directives used	
Directive 97/23 EC	"Safety of pressure equipment" (pressure equipment directive)
EN 982	Safety-related requirements on fluid power systems and their components Hydraulics
	Safety data sheets fluid suppliers

General technical data

Table 10: General technical data manifold

Type	Unloading	Pressure adjustment bar ¹⁾						Weight kg
ABZSS08	M	50	100	140	210	330	350	4.0
ABZSS10	M							5.2
	E							5.5
ABZSS20	M							8.5
	E							8.8
ABZSS30	M							20.5
	E							20.8
ABZSS30...SO30	M					315	---	26.5
	E							26.8
ABZSS-P30	M					330	350	33.1
	E							33.4

Type	Symbol	Pressure adjustment bar ¹⁾										Weight kg
0532VAW20	1	---	---	---	---	---	---	---	---	---	---	4.4
	2											4.7
	3											4.8
	4											5.6
	8											4.6
	10											4.5
0532VAW30	1	---	---	---	---	---	160	211	---	---	---	13.8
	2											14.3
	3											15.2
	4											14.7
	5						---	211	250	---	---	14.2
	7											14.4
	8											14.4
	9											14.3

¹⁾ Other pressure adjustments upon request.

Appendix

Hydraulics		
Hydraulic fluid temperature range	–15 °C to +80 °C	
Hydraulic fluid	Mineral oil (HL, HLP) according to DIN 51524 ¹⁾	
	Triglycerides (rape seed oil) HETG according to VDMA 24568 ¹⁾	
	Synthetic esters HEES according to VDMA 24568 ²⁾	
	Polyglycols HEPG according to VDMA 24568 ²⁾	
Cleanliness class according to ISO code	Maximum admissible degree of contamination of the hydraulic fluid according to ISO 4406 (c) class 20/18/15	
Maximum operating pressure	ABZSS10...	350 bar
	ABZSS20...	
	ABZSS30	
	ABZSS30...SO30	315 bar
	ABZSS-P30...	350 bar
	0532VAW...	330 bar

1) Suitable for FKM and NBR seals

2) Suitable for FKM seals

Electrical system	
Protection class according to VDE 0470-1 (DIN EN 60529) DIN 40050-9 - version "K4"	IP 65 with mating connector mounted and locked

16 Appendix

16.1 Address directory

Please refer to www.boschrexroth.com for addresses of foreign subsidiaries.

Appendix

Bosch Rexroth AG
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